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Cultural Biases in Equity Analysis

Contents

1	Overview	3
1.1	Research question	3
1.2	Core idea	3
1.3	Main findings	3
2	Data and construction of the key cultural-bias variable	3
2.1	Sample and objects	3
2.2	Measuring bilateral cultural trust bias	4
2.3	Other controls	4
3	All models	4
3.1	Baseline recommendation regression (Table II)	4
3.2	Time-varying effect of trust bias (Figure 1)	5
3.3	Sentiment interactions (Table IV)	5
3.4	Salience: eponymous firms and size (Table III)	5
3.5	Shocks to cultural perceptions: crisis and Brexit (Tables V–VI; Figures 2–3)	5
3.6	Announcement returns: market reaction to recommendations (Table VII)	5
3.7	Monthly returns: does trust bias reduce predictive content? (Table VIII)	6
3.8	Target prices (Table IX)	6
3.9	Earnings forecasts: forecast bias and absolute error (Table X)	6
3.10	Cross-sectional heterogeneity (Table XI)	6
4	Interpretation of all tables and figures	7
4.1	Table I: summary statistics (what the sample looks like)	7
4.2	Table II: trust bias predicts recommendation optimism	7
4.3	Table III: salience magnifies bias (eponymous firms and size)	8
4.4	Figure 1 and Table IV: bias strengthens in bad times	8
4.5	Table V and Figure 2: the European debt crisis generates a North–South wedge	9
4.6	Table VI and Figure 3: Brexit increases U.K. home bias	9
4.7	Table VII: announcement returns suggest markets discount biased buy recommendations	10
4.8	Table VIII: biased recommendations are worse predictors of monthly returns	10
4.9	Table IX: target prices are more optimistic when trust bias is higher	11
4.10	Table X: earnings forecasts are not more optimistic, and may be more accurate	11
4.11	Table XI: who is more biased? broker, analyst, and firm moderators	11

5	Short conclusion	13
6	Conclusion and intuition	13
6.1	Summary	13
6.2	Why these patterns arise	15
6.3	Takeaway	15

1 Overview

1.1 Research question

The paper asks whether **cultural perceptions between countries** (captured by survey-based bilateral trust) systematically bias **sell-side equity analysts’ outputs**—especially *stock recommendations*—in Europe, and whether such biases are *economically meaningful* for markets.

1.2 Core idea

Each analyst has a country of origin (analyst location / broker country), and each firm has a headquarter country. If analysts’ beliefs and judgments are affected by culturally shaped perceptions about “people from that country,” then recommendations and valuations may shift even after controlling for standard geography and investment-banking conflicts.

1.3 Main findings

1. **Recommendations are culturally biased.** Analysts are systematically more (less) positive when their home country has a more positive (negative) *cultural trust bias* toward the firm’s country (Table II).
2. **The bias strengthens when nationality is salient.** The trust-bias effect is stronger for *eponymous* “national champion” firms whose names include their country (and also increases with firm size) (Table III).
3. **The strength of bias varies over time with sentiment.** The trust-bias coefficient is larger in bad times: it co-moves positively with pessimism and negatively with consumer confidence (Figure 1 and Table IV).
4. **Short-term shocks to perceptions amplify cross-country divergence.** During the European debt crisis, northern analysts become more negative toward southern firms (Table V and Figure 2). Around Brexit, U.K. analysts become more home-biased relative to the rest of Europe (Table VI and Figure 3).
5. **Markets partly recognize biased recommendations, and bias reduces information content.** Buy recommendations from more positively biased analysts trigger weaker positive announcement reactions; for sell recommendations, the pattern reverses in a way consistent with stronger negative information content (Table VII). Moreover, biased recommendations are worse predictors of subsequent monthly returns, and conditioning on trust bias improves the return informativeness of the recommendation signal (Table VIII).
6. **Bias spills into target prices but not into (directional) earnings forecast bias.** More positively biased analysts set higher target prices (Table IX), while earnings forecast bias is not significantly higher; absolute forecast errors can even be smaller (Table X).

2 Data and construction of the key cultural-bias variable

2.1 Sample and objects

The empirical setting is European analysts and European firms. The main observation unit in the recommendation analysis is an *analyst–firm–month* observation (Table I).

2.2 Measuring bilateral cultural trust bias

The paper follows Guiso, Sapienza, and Zingales (2009) and uses Eurobarometer survey questions on trust between nationalities. Let $\text{Trust}_{i,j,t}$ denote how much citizens of origin country i report trusting citizens of destination country j in survey year t . To strip out mechanical components (some countries trust more in general; some are viewed as more trustworthy), estimate:

$$\text{Trust}_{i,j,t} = \lambda_i + \kappa_j + \gamma_t + \varepsilon_{i,j,t}. \quad (1)$$

Trust bias is defined as the residual:

$$\text{TrustBias}_{i,j} \equiv \hat{\varepsilon}_{i,j,t}.$$

Interpretation: $\text{TrustBias}_{i,j}$ captures the component of bilateral trust that differs from the “consensus” trust level once we control for (i) the origin country’s general tendency to trust, (ii) the destination country’s baseline perceived trustworthiness, and (iii) survey-year effects. By construction it is *relative* and *bilateral*.

2.3 Other controls

In the recommendation regressions the paper controls for:

- **Geography and home bias:** Same-country indicator; distance.
- **Investment-banking conflicts:** indicators for an IB relationship and for IB potential; plus broker market shares in syndicated loans, underwriting, and M&A advisory mandates.

3 All models

3.1 Baseline recommendation regression (Table II)

Let $\text{Rec}_{i,j,t}$ be analyst i ’s recommendation for firm j in month t (coded 1–5), and let $\text{TrustBias}_{c(i),c(j)}$ be trust bias between analyst-country $c(i)$ and firm-country $c(j)$. The baseline specification is:

$$\text{Rec}_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \beta \text{TrustBias}_{c(i),c(j)} + \phi' X_{i,j,t} + \varepsilon_{i,j,t}, \quad (2)$$

where:

- $\alpha_{i,t}$ are **analyst–month fixed effects** (absorbing analyst-specific optimism/pessimism in that month),
- $\gamma_{j,t}$ are **firm–month fixed effects** (absorbing firm-time news affecting all analysts in that month),
- $X_{i,j,t}$ are the controls listed above.

Identification intuition. Because both analyst–month and firm–month fixed effects are included, β is identified by **within-month cross-sectional variation** in which analysts cover which firms, after netting out (i) any analyst-specific time-varying tone and (ii) any firm-specific time-varying fundamentals/news.

3.2 Time-varying effect of trust bias (Figure 1)

To allow the trust-bias effect to vary by month, interact trust bias with month dummies:

$$\text{Rec}_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \sum_m \beta_m (\mathbb{1}\{t = m\} \cdot \text{TrustBias}_{c(i),c(j)}) + \phi' X_{i,j,t} + \varepsilon_{i,j,t}. \quad (3)$$

Figure 1 plots the estimated β_m against aggregate pessimism and consumer confidence (EU-level).

3.3 Sentiment interactions (Table IV)

To test whether the trust-bias effect strengthens in bad times, interact with sentiment proxies:

$$\text{Rec}_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \beta_1 (\text{Sentiment}_t \times \text{TrustBias}_{c(i),c(j)}) + \beta_0 \text{TrustBias}_{c(i),c(j)} + \phi' X_{i,j,t} + \varepsilon_{i,j,t}, \quad (4)$$

where Sentiment_t is either (i) **pessimism** or (ii) the **consumer confidence indicator**.

3.4 Salience: eponymous firms and size (Table III)

Let Eponymous_j indicate that the firm name includes the home-country name. The salience specification adds interactions:

$$\begin{aligned} \text{Rec}_{i,j,t} = & \alpha_{i,t} + \gamma_{j,t} + \beta \text{TrustBias}_{c(i),c(j)} + \theta (\text{Eponymous}_j \times \text{TrustBias}_{c(i),c(j)}) \\ & + \delta (\ln(\text{MktCap}_{j,t}) \times \text{TrustBias}_{c(i),c(j)}) + \phi' X_{i,j,t} + \varepsilon_{i,j,t}. \end{aligned} \quad (5)$$

3.5 Shocks to cultural perceptions: crisis and Brexit (Tables V–VI; Figures 2–3)

European debt crisis. Let Crisis_t indicate the Eurozone recession window, and define: NorthernAnalyst_i and SouthernFirm_j based on country groupings. The key triple-difference term is:

$$\text{Rec}_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \beta (\text{Crisis}_t \times \text{NorthernAnalyst}_i \times \text{SouthernFirm}_j) + \dots \quad (6)$$

Figure 2 complements this with a coefficient path over time for the interaction.

Brexit. Let Post_t indicate the post-Article-50 period and define BritishAnalyst_i . The key term is:

$$\text{Rec}_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \beta (\text{Post}_t \times \text{BritishAnalyst}_i \times \text{SameCountry}_{i,j}) + \dots \quad (7)$$

Figure 3 plots the time series of the British home-bias interaction coefficient.

3.6 Announcement returns: market reaction to recommendations (Table VII)

Let $\text{CAR}_{i,j,t}$ be the 3-day cumulative abnormal return around the recommendation announcement (days -1 to $+1$), computed using a local Fama–French 4-factor model. The regression is:

$$\text{CAR}_{i,j,t} = \alpha_i + \gamma_j + \lambda_t + \beta \text{TrustBias}_{c(i),c(j)} + \phi' X_{i,j,t} + \varepsilon_{i,j,t}. \quad (8)$$

The paper estimates (8) separately for buy-type and sell-type events.

3.7 Monthly returns: does trust bias reduce predictive content? (Table VIII)

Stocks are sorted each month into recommendation quintiles (country-specific cutoffs). Let $\text{ExRet}_{j,t}$ be the monthly excess stock return. Let $\text{RecQ}_{j,t}$ be recommendation-quintile dummies. Let $\text{TrustBias}_{j,t}$ be the *average* trust bias across sample analysts covering stock j in month t . The specification is:

$$\text{ExRet}_{j,t} = \alpha_0 + \sum_{q=2}^5 \gamma_q \mathbb{1}\{\text{RecQ}_{j,t} = q\} + \sum_{q=1}^5 \beta_q \left(\mathbb{1}\{\text{RecQ}_{j,t} = q\} \cdot \text{TrustBias}_{j,t} \right) + \psi' Z_{j,t} + u_{j,t}, \quad (9)$$

where $Z_{j,t}$ includes standard return predictors and/or fixed effects (country-month FE in some columns).

3.8 Target prices (Table IX)

Let $\text{TP}_{i,j,t}$ be the latest target price for analyst–firm pair scaled by beginning-of-month price. The target-price regression mirrors the recommendation framework (monthly panel with analyst–month and firm–month FE):

$$\text{TP}_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \beta \text{TrustBias}_{c(i),c(j)} + \phi' X_{i,j,t} + \varepsilon_{i,j,t}, \quad (10)$$

estimated under different maximum “age” cutoffs for target-price staleness.

3.9 Earnings forecasts: forecast bias and absolute error (Table X)

Define (at the fiscal-year level):

$$\text{FB}_{i,j,t} = \text{EPSestimate}_{i,j,t} - \text{ActualEPS}_{j,t}, \quad (11)$$

$$\text{PMFB}_{i,j,t} = \frac{\text{FB}_{i,j,t} - \text{MFB}_{j,t}}{|\text{MFB}_{j,t}|}, \quad (12)$$

$$\text{PMAFE}_{i,j,t} = \frac{\text{AFE}_{i,j,t} - \text{MAFE}_{j,t}}{\text{MAFE}_{j,t}}, \quad (13)$$

where $\text{MFB}_{j,t}$ and $\text{MAFE}_{j,t}$ are consensus mean forecast bias and consensus mean absolute forecast error, respectively. The regression is:

$$Y_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \beta \text{TrustBias}_{c(i),c(j)} + \phi' X_{i,j,t} + \varepsilon_{i,j,t}, \quad (14)$$

with $Y \in \{\text{PMFB}, \text{PMAFE}\}$ and alternative fixed-effect structures (analyst/firm/year vs analyst-year/firm-year).

3.10 Cross-sectional heterogeneity (Table XI)

To test moderators, the paper estimates interaction regressions of the form:

$$\text{Rec}_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \beta \text{TrustBias}_{c(i),c(j)} + \theta \left(M_{i,j,t} \times \text{TrustBias}_{c(i),c(j)} \right) + \phi' X_{i,j,t} + \varepsilon_{i,j,t}, \quad (15)$$

where $M_{i,j,t}$ is a broker characteristic (diversity/size), an analyst characteristic (experience/coverage length/attitudes), or a firm characteristic (size/opacity proxies).

4 Interpretation of all tables and figures

4.1 Table I: summary statistics (what the sample looks like)

Table I documents the scale and composition of the analyst–firm–month panel:

- Recommendations are on average slightly above “Hold”: mean recommendation ≈ 3.53 (median = 4), consistent with a positive bias in sell-side outputs.
- The buy-recommendation indicator has mean ≈ 0.51 , while sell recommendations are less frequent (mean ≈ 0.17).
- Trust bias has mean ≈ 0.22 and standard deviation ≈ 0.15 .
- The sample is heavily domestic: SameCountry = 1 about 74% of the time.
- Eponymous firms are rare but nontrivial: about 6.6% of observations.
- IB relationships are present for a meaningful share of pairs: IB relationship mean ≈ 0.12 ; IB potential mean ≈ 0.28 .

Why this matters: there is large cross-sectional variation in trust bias and in domestic vs foreign coverage, which supports identification with fixed effects, while the IB and geography variables are nontrivial and need controls.

Table I
Summary Statistics
This table presents summary statistics for the analyst-firm-month observations in the sample. The sample period is 1996 to 2018. *Recommendation* is the analyst recommendation, coded from 1 (lowest, *Strong sell*) to 5 (highest, *Strong buy*). *Buy recommendation* is a dummy taking the value of one if the recommendation is 5 (*Strong buy*) or 4 (*Buy*). *Sell recommendation* is a dummy taking the value of one if the recommendation is 1 (*Strong sell*) or 2 (*Sell*). All variables are defined in the [Appendix](#).

	Mean	Std	p10	p50	p90
Recommendations					
Recommendation	3.528	1.151	2.000	4.000	5.000
Buy rec.	0.508	0.500	0.000	1.000	1.000
Sell rec.	0.189	0.375	0.000	0.000	1.000
Trust bias and geography					
Trust bias	0.218	0.148	-0.012	0.246	0.387
Same country	0.738	0.440	0.000	1.000	1.000
Distance ('000 km)	0.592	1.354	0.000	0.194	0.841
Broker					
Broker IHH	0.464	0.524	0.182	0.433	0.792
Broker nationalities	8.382	7.002	2.000	6.000	20.000
Broker size	46.014	51.895	8.000	25.000	112.000
Top 10	0.394	0.460	0.000	0.000	1.000
Analyst					
Analyst <i>N</i> firms	11.534	8.275	4.000	10.000	20.000
Years covered	3.608	3.235	0.318	1.912	7.263
Ans. experience (yrs)	5.834	4.769	1.036	4.419	12.934
Anti-globalization	0.441	0.135	0.299	0.393	0.651
Firm					
<i>N</i> recommendations	16.408	11.707	3.000	14.000	33.000
<i>N</i> rec. (in sample)	5.803	4.341	1.000	5.000	12.000
Eponymous	0.066	0.249	0.000	0.000	0.000
Market cap (USDbl)	9.155	20.768	0.097	1.493	26.883
Inst. ownership	0.209	0.143	0.034	0.187	0.416
IVOL	0.072	0.034	0.039	0.065	0.113
Mgmt guidance	0.209	0.406	0.000	0.000	1.000
Broker relationship					
IB relationship	0.119	0.324	0.000	0.000	1.000
IB potential	0.281	0.449	0.000	0.000	1.000
Has IB	0.681	0.466	0.000	1.000	1.000
IB client	0.244	0.475	0.000	0.000	1.000
Share - syndicated loans	0.012	0.064	0.000	0.000	0.000
Share - underwriting	0.020	0.107	0.000	0.000	0.000
Share - advisory	0.008	0.070	0.000	0.000	0.000
<i>N</i>	1,269,560				

Table II
Recommendations and Trust Bias

This table presents regression results for analyst recommendations. The dependent variable is shown above each column. *Recommendation* is the analyst recommendation, coded from 1 (lowest, *Strong sell*) to 5 (highest, *Strong buy*). *Buy recommendation* is a dummy taking the value of one if the recommendation is 5 (*Strong buy*) or 4 (*Buy*). *Sell recommendation* is a dummy taking the value of one if the recommendation is 1 (*Strong sell*) or 2 (*Sell*). The sample period is 1996 to 2018. Variables are defined in the [Appendix](#). Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and month, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

Panel A: All analysts				
	Rec. (1-5)	Buy rec.	Sell rec.	
	(1)	(2)	(3)	(4)
Trust bias	0.9583*** (0.0873)	0.5910*** (0.1514)	0.2802*** (0.0724)	-0.1421*** (0.0460)
Same country	0.0702 (0.0488)	0.0244 (0.0238)	-0.0151 (0.0141)	
ln(Distance)	-0.0158*** (0.0053)	-0.0078*** (0.0016)	0.0056*** (0.0010)	
IB relationship	0.0763*** (0.0139)	0.0302*** (0.0050)	-0.0323*** (0.0083)	
IB potential	0.0510 (0.0273)	0.0094 (0.0095)	-0.0088 (0.0073)	
Share - syndicated loans	-0.0506 (0.0761)	-0.0324 (0.0377)	0.0331 (0.0207)	
Share - underwriting	0.1166 (0.0996)	0.0708 (0.0447)	-0.0105 (0.0199)	
Share - advisory	0.1779*** (0.0633)	0.0912*** (0.0380)	-0.0299 (0.0185)	
Firm-Month FE	Yes	Yes	Yes	
Analyst-Month FE	Yes	Yes	Yes	
<i>N</i>	1,035,166	1,035,166	1,035,166	1,035,166
<i>R</i> ²	0.564	0.565	0.547	0.547
Panel B: Excluding domestic analysts				
	Rec. (1-5)	Buy rec.	Sell rec.	
	(1)	(2)	(3)	(4)
Trust bias	0.7811** (0.3429)	0.6883* (0.3496)	0.2576* (0.1453)	-0.2086** (0.0959)
ln(Distance)	-0.0268*** (0.0093)	-0.0115*** (0.0044)	0.0103*** (0.0020)	
IB relationship	0.1396** (0.0530)	0.0481* (0.0248)	-0.0712*** (0.0168)	
IB potential	0.0406 (0.0619)	-0.0005 (0.0280)	0.0073 (0.0188)	
Share - syndicated loans	-0.0712 (0.2391)	-0.0286 (0.0809)	0.0254 (0.0814)	

Panel B: Excluding domestic analysts				
	Rec. (1-5)	Buy rec.	Sell rec.	
	(1)	(2)	(3)	(4)
Share - underwriting		0.5384** (0.2353)	0.2787*** (0.1176)	0.0005 (0.0595)
Share - advisory		0.2406 (0.1518)	0.1491*** (0.0710)	-0.0345 (0.0465)
Firm-Month FE	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes
<i>N</i>	173,274	173,274	173,274	173,274
<i>R</i> ²	0.683	0.684	0.685	0.673

4.2 Table II: trust bias predicts recommendation optimism

Table II is the main “existence” result.

Panel A (all analysts).

- The trust-bias coefficient is positive and highly significant for the 1–5 recommendation score. With controls, the estimate is about 0.59 (s.e. ≈ 0.15).

- In a linear-probability setting, trust bias increases the probability of a buy recommendation: the coefficient is about 0.28 (s.e. ≈ 0.07).
- Trust bias decreases the probability of a sell recommendation: coefficient about -0.14 (s.e. ≈ 0.05).

Economic meaning: a more positive cultural perception toward the firm’s country translates into systematically more optimistic calls, even *within the same month* for the same analyst and the same firm.

Panel B (excluding domestic analyst–firm pairs). The trust-bias effect remains positive for recommendation levels and buy propensity (and negative for sells), showing the relationship is not mechanically driven by domestic coverage.

Table III
Eponymous Firms and the Salience of Nationality

This table presents regression results for analyst recommendations. The dependent variable is shown above each column. *Recommendation* is the analyst recommendation, coded from 1 (lowest, *Strong sell*) to 5 (highest, *Strong buy*). *Buy recommendation* is a dummy taking the value of one if the recommendation is 5 (*Strong buy*) or 4 (*Buy*). *Eponymus* is a dummy taking the value of one if the firm name includes the name of its home country. Controls include *Same country*, *ln(Distance)*, *IB relationship*, *IB potential*, *Share - syndicated loans*, *Share - underwriting*, and *Share - advisory*. The sample period is 1996 to 2018. Variables are defined in the Appendix. Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and month, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

	Rec. (1–5)					Buy rec.
	(1)	(2)	(3)	(4)	(5)	(6)
Eponymous $\times$ Trust bias	0.2208** (0.1325)	0.3188** (0.1354)	0.2659* (0.1475)	0.1170** (0.0472)	0.1198** (0.0512)	0.1052* (0.0552)
ln(Market cap) $\times$ Trust bias			0.0963*** (0.0256)		0.0271*** (0.0104)	
Trust bias	0.5336*** (0.1480)			0.3608*** (0.0725)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-pair FE	No	Yes	No	Yes	No	Yes
N	1,035,166	1,035,166	1,035,166	1,035,166	1,035,166	1,035,166
R ²	0.565	0.566	0.568	0.547	0.549	0.550

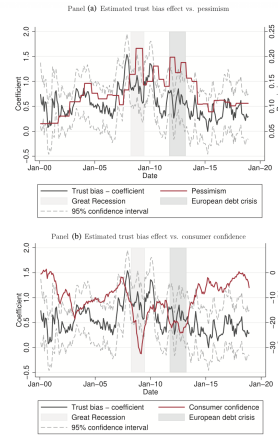


Figure 1. Trust bias versus pessimism and consumer confidence. This figure plots monthly estimates of regression coefficients for *Trust bias* from the regression below against the aggregate level of pessimism and consumer confidence in the European Union. Regression equation:
 $Recommendation_{i,j,t} = \alpha_{i,j} + \gamma_{j,t} + \beta Month_{i,j} \times Trust\ bias_{i,j} + \phi K_{i,j} + \epsilon_{i,j,t}$.

Table IV
The Effect of Cultural Bias versus Sentiment

This table presents regression results for analyst recommendations. The dependent variable is shown above each column. *Recommendation* is the analyst recommendation, coded from 1 (lowest, *Strong sell*) to 5 (highest, *Strong buy*). *Buy recommendation* is a dummy taking the value of one if the recommendation is 5 (*Strong buy*) or 4 (*Buy*). *Pessimism* is the proportion of people who expect their life to be worse in the next year, based on Eurobarometer surveys. *Consumer confidence* is the Consumer Confidence Indicator for the EU, published by the European Commission. Controls include *Same country*, *ln(Distance)*, *IB relationship*, *IB potential*, *Share - syndicated loans*, *Share - underwriting*, and *Share - advisory*. The sample period is 1996 to 2018. Variables are defined in the Appendix. Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and month, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

	Rec. (1–5)					Buy rec.
	(1)	(2)	(3)	(4)	(5)	(6)
Pessimism $\times$ Trust bias	2.9518*** (0.3987)	3.4083*** (0.9064)	3.3153*** (1.2830)	0.8864*** (0.3977)	1.1468*** (0.4000)	1.3807*** (0.5538)
Trust bias	0.2179 (0.2099)			0.1795* (0.0573)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-pair FE	No	Yes	No	No	Yes	No
Analyst-Firm FE	No	No	Yes	No	No	Yes
N	1,023,849	1,023,849	1,023,849	1,023,849	1,023,849	1,023,849
R ²	0.565	0.566	0.568	0.547	0.549	0.550

	Rec. (1–5)					Buy rec.
	(1)	(2)	(3)	(4)	(5)	(6)
CCI $\times$ Trust bias	-0.0155*** (0.0046)	-0.0169*** (0.0046)	-0.0164*** (0.0060)	-0.0045*** (0.0020)	-0.0053*** (0.0019)	-0.0083*** (0.0028)
Trust bias	0.4135*** (0.1718)			0.2286*** (0.0739)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-pair FE	No	Yes	No	No	Yes	No
Analyst-Firm FE	No	No	Yes	No	No	Yes
N	1,035,166	1,035,166	1,035,166	1,035,166	1,035,166	1,035,166
R ²	0.565	0.566	0.568	0.547	0.549	0.550

4.3 Table III: salience magnifies bias (eponymous firms and size)

Table III tests the prediction that cultural cues matter more when nationality is more salient.

- The interaction Eponymous $\times$ TrustBias is positive and significant (recommendation columns: about 0.27–0.32; buy columns: about 0.105–0.12).
- The interaction ln(MktCap) $\times$ TrustBias is also positive and significant (about 0.10 for recommendations and about 0.03 for buys).

Interpretation: cultural bias is not uniform. It intensifies when the firm is more “nationally identifiable” and for larger firms, consistent with an attention/salience mechanism.

4.4 Figure 1 and Table IV: bias strengthens in bad times

Figure 1. Monthly estimates of the trust-bias slope co-move with sentiment: the slope is higher when pessimism is high and when consumer confidence is low, with recessions highlighted. This suggests cultural biases become more influential under stress.

Table IV.

- In Panel A, the interaction $\text{Pessimism}_t \times \text{TrustBias}$ is strongly positive (recommendation columns ≈ 3.0 – 3.4 ; buy columns ≈ 0.9 – 1.4).
- In Panel B, the interaction $\text{CCI}_t \times \text{TrustBias}$ is negative and significant (recommendation columns around -0.016 ; buy around -0.005).

Interpretation: the same cultural cross-section matters more when aggregate sentiment is worse. This supports a “stress amplifies bias” mechanism rather than a purely time-invariant information story.

4.5 Table V and Figure 2: the European debt crisis generates a North–South wedge

Table V (Panel A). The triple interaction $\text{Crisis} \times \text{NorthernAnalyst} \times \text{SouthernFirm}$ is negative and significant: recommendations become more negative (about -0.37 to -0.43), and the probability of a buy recommendation falls (about -0.23 to -0.25).

Table V
European Debt Crisis and North-South Biases

This table presents regression results for analyst recommendations. The dependent variable is shown above each column. Recommendation is the analyst recommendation, scaled from 1 (strong buy) to 5 (strong sell). Buy recommendation is a dummy taking the value of one if the recommendation is 1 (Strong buy) or 4 (Buy). Crisis is a dummy taking the value of one during the Recession months of Q4 2011 to Q1 2013. Northern analyst is a dummy taking the value one if the analyst is from Germany, the United Kingdom, Netherlands, Austria, Sweden, Denmark, or Finland. Southern firm is a dummy taking the value one if the firm is from Portugal, Italy, Greece, or Spain. Controls include Size, country, industry, B/M ratio, momentum, share, concentration, share, ownership, and share, industry. The sample period is 2008 to 2014. Variables are defined in the Appendix. Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and month, are shown in parentheses. Significance levels: *1%, **5%, ***10%.

	Panel A: Northern analyst and Southern firm			
	Rec. (1-5)		Buy rec.	
	(1)	(2)	(3)	(4)
Crisis × Northern s. × Southern f.	-0.3749*** (0.1179)	-0.4237*** (0.1393)	-0.2311*** (0.1140)	-0.2480*** (0.0810)
Northern s. × Southern f.	-0.0460 (0.0460)	-0.0460 (0.0460)	-0.0460 (0.0460)	-0.0460 (0.0460)
Controls	Yes	Yes	Yes	Yes
First Month FE	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes
Country-pair FE	Yes	Yes	Yes	Yes
Analyst-Firm FE	No	No	No	No
N	302,547	302,547	302,547	302,547
R ²	0.185	0.189	0.185	0.185

Figure 1: Caption 1

Table V (Panel B). The opposite direction (southern analysts toward northern firms) is smaller and mostly statistically weaker.

Figure 2. The time-series coefficient for the north-on-south interaction turns sharply negative around the crisis window.

Interpretation: this is evidence for *short-horizon shifts in cultural perceptions* that cannot be captured by a static trust measure alone. The crisis triggered a perception shock consistent with a narratives-driven “blame” channel.

4.6 Table VI and Figure 3: Brexit increases U.K. home bias

Table VI shows that after Article 50, U.K. analysts become more home-biased:

- Post × British × SameCountry is positive and highly significant (recommendations about 0.48–0.70; buy propensity about 0.17–0.34).

Figure 3 visualizes this divergence over time (with markers for the referendum and Article 50).

Interpretation: a major political shock that affects national identity and cross-border relations is mirrored in analysts’ home bias.

Table V—Continued

Panel B: Southern analyst and Northern firm

	Rec. (1-5)		Buy rec.	
	(1)	(2)	(3)	(4)
Crisis × Southern s. × Northern f.	-0.1203 (0.1109)	-0.1406 (0.1054)	-0.1313 (0.1031)	-0.0804 (0.0486)
Southern s. × Northern f.	-0.0389 (0.0447)	-0.0389 (0.0447)	-0.0389 (0.0447)	-0.0389 (0.0447)
Controls	Yes	Yes	Yes	Yes
First Month FE	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes
Country-pair FE	Yes	Yes	Yes	Yes
Analyst-Firm FE	No	No	No	No
N	302,547	302,547	302,547	302,547
R ²	0.185	0.189	0.185	0.185

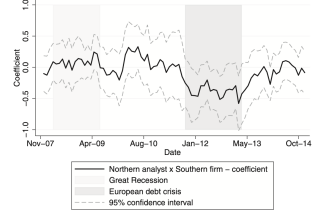


Figure 2. European debt crisis, northern analysts, and southern firms. This figure plots monthly estimates of regression coefficients for Northern analyst × Southern firm from the regression

$$\text{Recommendation}_{i,j,t} = \alpha_1 + \gamma_{1,t} + \beta \text{Month}_t \times \text{Northern analyst}_i \times \text{Southern firm}_j + \alpha_2 \text{Month}_t + \gamma_{2,t}$$

Table VI
Brexit and United Kingdom versus the Rest of Europe

This table presents regression results for analyst recommendations. The dependent variable is shown above each column. *Recommendation* is the analyst recommendation, coded from 1 (lowest, *Strong sell*) to 5 (highest, *Strong buy*). *Buy recommendation* is a dummy taking the value of one if the recommendation is 4 (*Strong buy*) or 5 (*Buy*). *Post* is a dummy taking the value of one after March 29, 2017, which is when the United Kingdom invoked Article 50, formally beginning the Brexit process. Controls include *Same country*, *ln(Distance)*, *IB relationship*, *IB potential*, *Share* - syndicated loans, *Share* - underwriting, and *Share* - advisory. The sample period is 2015 to 2018. Variables are defined in the Appendix. Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and month, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

	Rec. (1-5)			Buy rec.		
	(1)	(2)	(3)	(4)	(5)	(6)
Post × British × Same country	0.7002*** (0.1484)	0.6886*** (0.1582)	0.4778*** (0.1235)	0.3217*** (0.0639)	0.3357*** (0.0711)	0.1685** (0.0640)
Post × Same country	-0.0721** (0.0387)	-0.0885** (0.0353)	0.0270 (0.0356)	-0.0115 (0.0173)	-0.0284 (0.0171)	0.0244 (0.0197)
British × Same country	-0.3638** (0.1480)			-0.0388 (0.0794)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-pair FE	No	Yes	No	No	Yes	No
Analyst-Firm FE	No	No	Yes	No	No	Yes
N	147,331	147,331	147,359	147,331	147,331	147,359
R ²	0.570	0.578	0.830	0.553	0.562	0.817

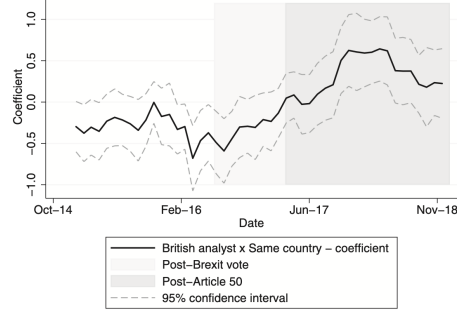


Figure 3. Brexit and United Kingdom versus the rest of Europe. This figure plots monthly estimates of regression coefficients for *British analyst × Same country* from the regression

$$\text{Recommendation}_{i,j,t} = \alpha_{i,t} + \gamma_{j,t} + \beta \text{Month}_t \times \text{British analyst}_i \times \text{Same country}_{i,j} + \psi \text{Month}_t \times \text{Same country}_{i,j} + \phi X_{i,j,t} + \epsilon_{i,j,t}$$

4.7 Table VII: announcement returns suggest markets discount biased buy recommendations

Table VII links trust bias to the market reaction around recommendation events.

- **Buy recommendations:** trust bias enters negatively (about -0.014 to -0.018), meaning that a more positively biased analyst triggers *weaker positive* CARs. This is consistent with the market treating these buy calls as less informative.
- **Sell recommendations:** effects are more visible for active analysts and for downgrades to sell: trust bias is negative (about -0.03 to -0.04), implying *more negative* CARs. This is consistent with the idea that sell calls from generally “optimistic” analysts are more surprising and thus more informative.

Table VII
Recommendation Announcement Returns

This table presents regression results for recommendation announcement returns. The dependent variable in Panels B and C is the three-day cumulative abnormal return over days -1 to 1 relative to the recommendation announcement day. Abnormal returns are estimated based on the Fama-French four-factor model at the country level and factor betas are estimated from daily returns during trading days $(-252, -42)$ relative to the event date. Controls include *Same country*, *ln(Distance)*, *IB relationship*, *IB potential*, *Share* - syndicated loans, *Share* - underwriting, and *Share* - advisory. The sample period is 1996 to 2018. Variables are defined in the Appendix. Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and announcement date, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

Panel A: Summary statistics	Mean	Std	p10	p50	p90
CAR	-0.000	0.038	-0.043	-0.001	0.045
CAR - Buy	0.005	0.038	-0.037	0.003	0.052
CAR - Sell	-0.007	0.040	-0.053	-0.006	0.038
CAR - Upgrade to buy	0.008	0.038	-0.032	0.006	0.037
CAR - Downgrade to sell	-0.009	0.041	-0.057	-0.008	0.037
Trust bias	0.218	0.149	-0.018	0.246	0.387
N	92,084				

Panel B: Buy recommendations					
	Buy (all)		Buy (active analysts)		Upgrade to buy
	(1)	(2)	(3)	(4)	(5)
Trust bias	-0.0135** (0.0055)	-0.0142** (0.0055)	-0.0179** (0.0073)	-0.0167** (0.0076)	-0.0166* (0.0091)
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Analyst FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes
Year-Month FE	No	Yes	No	Yes	No
N	42,600	42,600	27,487	27,487	18,269
R ²	0.182	0.190	0.210	0.223	0.275

Panel C: Sell recommendations					
	Sell (all)		Sell (active analysts)		Downgrade to sell
	(1)	(2)	(3)	(4)	(5)
Trust bias	-0.0046 (0.0101)	-0.0033 (0.0096)	-0.0304* (0.0159)	-0.0292* (0.0165)	-0.0387** (0.0181)
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Analyst FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes
Year-Month FE	No	Yes	No	Yes	No
N	15,239	15,239	10,684	10,684	8,144
R ²	0.277	0.293	0.315	0.336	0.353

Table VIII
Monthly Stock Returns and Trust Bias

This table presents regression results for monthly stock returns. Panel A presents summary statistics for the monthly stock returns sample. The dependent variable in Panel B is *Excess return*, the monthly stock return less the risk-free rate. *Return* is the monthly raw stock return. *Trust bias* is calculated as the average trust bias across all sample analysts covering the stock, on a monthly basis. *Market beta* is calculated using a rolling 12-month window based on monthly returns. *Recommendation quintile* is calculated based on the average recommendation of the sample analysts, on a monthly basis, with limits set by firm country. The sample period is 1996 to 2018. Heteroskedasticity-consistent standard errors, clustered by firm, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

Panel A: Summary statistics					
	Mean	Std	p10	p50	p90
Stock return	0.432	10.516	-11.227	0.180	12.065
Excess return	0.067	10.505	-11.062	0.300	12.205
Return	11.235	43.456	-38.394	8.133	60.056
Return (LTM)					
Trust bias and controls					
Trust bias	0.219	0.124	0.044	0.246	0.387
ln(Market cap.)	6.604	2.052	3.918	6.539	9.386
R/M	0.780	0.838	0.176	0.558	1.539
RoE	0.089	2.200	-0.114	0.102	0.265
Market beta	0.852	0.834	-0.023	0.794	1.804
N	291,839				

Panel B: Monthly excess return regressions					
	(1)	(2)	(3)	(4)	(5)
Rec. quintile 2	0.4370*** (0.0644)	0.4142*** (0.0712)	0.3865*** (0.0621)	0.3453*** (0.1262)	0.4239*** (0.1396)
Rec. quintile 3	0.4176*** (0.0641)	0.3503*** (0.0601)	0.4163*** (0.0591)	-0.0208 (0.1190)	0.0038 (0.1348)
Rec. quintile 4	0.5439*** (0.0647)	0.5037*** (0.0722)	0.5137*** (0.0630)	0.1400 (0.1386)	0.1741 (0.1488)
Rec. quintile 5	0.5113*** (0.0762)	0.4949*** (0.0875)	0.5809*** (0.0758)	0.4862*** (0.1629)	0.3540** (0.1790)
Rec. quintile 1 × Trust bias				-1.4534*** (0.3502)	-1.1498*** (0.4152)
Rec. quintile 2 × Trust bias				-0.9680** (0.4016)	-1.1277** (0.4454)
Rec. quintile 3 × Trust bias				0.6431* (0.3824)	0.5578 (0.4388)
Rec. quintile 4 × Trust bias				0.4384 (0.4210)	0.3922 (0.4302)
Rec. quintile 5 × Trust bias				-1.1669** (0.6776)	-0.3436 (0.5761)
Market beta	-0.0162 (0.0316)	0.0091 (0.0291)		-0.0158 (0.0316)	0.0087 (0.0292)
ln(Market cap.)	-0.0188 (0.0119)	0.027** (0.0110)		-0.0252** (0.0123)	0.0288** (0.0115)

Table VIII—Continued

Panel B: Monthly excess return regressions					
	(1)	(2)	(3)	(4)	(5)
R/M	0.7363*** (0.0581)	0.4033*** (0.0517)		0.7359*** (0.0582)	0.4049*** (0.0517)
RoE	0.0239*** (0.0074)	0.0188*** (0.0085)		0.0234*** (0.0074)	0.0184** (0.0087)
Return (t-1)	0.0430*** (0.0033)	-0.0275*** (0.0038)		0.0429*** (0.0033)	-0.0276*** (0.0038)
Return (LTM)	0.0111*** (0.0006)	0.0109*** (0.0006)		0.0111*** (0.0006)	0.0108*** (0.0008)
Constant	0.0885** (0.0441)	-0.2518** (0.1149)		0.3935*** (0.0834)	0.0319 (0.1486)
Country-Month FE	No	No	Yes	No	Yes
N	290,872	194,860	194,684	290,872	194,684
R ²	0.000	0.007	0.293	0.001	0.007

4.8 Table VIII: biased recommendations are worse predictors of monthly returns

Table VIII asks whether trust bias reduces the predictive power of recommendation signals. The logic is: if high trust bias inflates buys and delays sells, then extreme recommendations by high-bias

analysts are less informative.

The interaction pattern is consistent with this:

- In the most negative recommendation quintiles, the interaction with trust bias is strongly negative (e.g., quintile 1 $\times$ trust bias around -1.43 in one specification), suggesting that high-bias “sell-side pessimism” predicts lower returns (the sell signal is more credible when issued by an otherwise optimistic/bias-prone analyst).
- In the top recommendation quintile, the interaction is also negative in some specifications (e.g., quintile 5 $\times$ trust bias around -1.17), consistent with high-bias buy signals being over-optimistic and thus less useful.
- Middle quintiles show offsetting positive interactions, mechanically required if bias reduces informativeness at both extremes.

Practical implication emphasized in the text: conditioning on trust bias improves the informativeness of recommendation-based trading strategies compared to sorting on recommendations alone.

4.9 Table IX: target prices are more optimistic when trust bias is higher

Table IX shows a clear and robust pattern:

- Trust bias increases scaled target prices across all staleness cutoffs.
- The effect is larger when focusing on *more recent* target prices (e.g., stronger for a 30-day maximum age cutoff than for 360 days), consistent with the bias being reflected in actively maintained valuations rather than only in stale targets.

4.10 Table X: earnings forecasts are not more optimistic, and may be more accurate

Table X is important because it separates “soft” analyst outputs (recommendations/targets) from more accountable ones (EPS forecasts):

- For **directional forecast bias** (PMFB), the trust-bias coefficient is not statistically significant across specifications.
- For **absolute forecast error** (PMAFE), the trust-bias coefficient is negative and significant (e.g., about -0.13 to -0.26), indicating *smaller* absolute errors for more positively biased analysts.

Interpretation: cultural bias appears to influence qualitative judgments (recommendations and target valuations), but it does not translate into mechanically more optimistic EPS numbers, likely because EPS forecast errors are easily audited ex post.

4.11 Table XI: who is more biased? broker, analyst, and firm moderators

Table XI decomposes where bias is stronger or weaker.

Table IX
Target Price

This table presents regression results for analyst target prices. The dependent variable in Panel B is *Target price*, the most recent broker target price divided by beginning-of-month stock price. The analysis is performed using monthly observations, for several maximum allowed target price ages, calculated as days from the target price announcement. The sample period is 1999 to 2018 (IBES PTG data begin in 1999 for my sample firms). Variables are defined in the [Appendix](#). Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and month, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

Panel A: Summary statistics					
	Mean	Std	p10	p50	p90
Target price (360)	1.168	0.313	0.874	1.111	1.485
Target price (30)	1.147	0.255	0.891	1.116	1.399
Trust bias	0.218	0.152	-0.015	0.246	0.387
<i>N</i>	623,833				
Panel B: Target prices and trust bias					
	Maximum target price age				
	(1) 360	(2) 180	(3) 90	(4) 60	(5) 30
Trust bias	0.0825** (0.0320)	0.1043*** (0.0306)	0.1300*** (0.0303)	0.1425*** (0.0341)	0.2368*** (0.0752)
Same country	0.0004 (0.0092)	-0.0025 (0.0093)	-0.0103 (0.0096)	-0.0171 (0.0112)	-0.0518** (0.0219)
ln(Distance)	-0.0017 (0.0011)	-0.0011 (0.0013)	-0.0011 (0.0014)	-0.0015 (0.0015)	-0.0028** (0.0011)
IB relationship	0.0075** (0.0036)	0.0087*** (0.0027)	0.0072** (0.0032)	0.0045 (0.0041)	0.0035 (0.0064)
IB potential	0.0182*** (0.0050)	0.171*** (0.0044)	0.0176*** (0.0052)	0.0204*** (0.0056)	0.0093 (0.0069)
Share - syndicated loans	0.0028 (0.0287)	-0.0075 (0.0241)	-0.0005 (0.0243)	0.0233 (0.0241)	0.0516 (0.0357)
Share - underwriting	0.0220 (0.0170)	0.0162 (0.0209)	0.0011 (0.0204)	0.0035 (0.0223)	0.0004 (0.0232)
Share - advisory	0.0224** (0.0102)	0.0192** (0.0088)	0.0132 (0.0120)	0.0223 (0.0137)	0.0177 (0.0117)
Firm-Month FE	Yes	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	496,069	376,915	230,952	144,853	47,723
<i>R</i> ²	0.757	0.778	0.810	0.832	0.874

Table X
Forecast Bias and Error

This table presents regression results for analyst forecast bias and forecast errors. This analysis is run using a yearly panel of observations with the most recent EPS estimate for each analyst-firm pair at the fiscal year end included. The dependent variable in Panel B is shown above each column. *PMFB* is calculated as the analyst forecast bias (EPS estimate less the actual reported EPS) less the consensus mean forecast bias, scaled by the absolute value of the consensus mean forecast bias. *PMAFE* is calculated as the absolute forecast error (absolute value of EPS estimate less the actual reported EPS) less the consensus mean absolute forecast error, scaled by the consensus mean absolute forecast error. *ln(Days to FY end - ann.)* is the number of days from the announcement day of the earnings forecast to the fiscal year-end. The sample period is 1996 to 2018. Variables are defined in the [Appendix](#). Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and year, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

	Panel A: Summary statistics					
	Mean	Std	p10	p50	p90	
PMFB	-0.019	1.599	-1.835	0.000	1.807	
PMAFE	-0.037	0.583	-0.802	-0.022	0.832	
Trust bias	0.216	0.150	-0.018	0.246	0.387	
<i>N</i>	101,894					
Panel B: Forecast errors and trust bias						
	PMFB			PMAFE		
	(1)	(2)	(3)	(4)	(5)	(6)
Trust bias	0.0450 (0.1656)	0.0393 (0.1690)	0.2247 (0.2069)	-0.1308** (0.0587)	-0.1350** (0.0587)	-0.2565** (0.0917)
Same country	-0.0097 (0.0597)	-0.0059 (0.0607)	-0.0297 (0.0779)	-0.0026 (0.0166)	0.0002 (0.0169)	0.0087 (0.0257)
ln(Days to FY end - ann.)	0.1153*** (0.0214)	0.1117*** (0.0212)	0.1271*** (0.0218)	0.0869*** (0.0071)	0.0866*** (0.0072)	0.0930*** (0.0080)
ln(Distance)	0.0052 (0.0042)	0.0062 (0.0040)	0.0085 (0.0072)	0.0005 (0.0016)	0.0007 (0.0017)	0.0007 (0.0028)
IB relationship	0.0158 (0.0441)	0.0041 (0.0439)	0.0143 (0.0454)	0.0150 (0.0121)	0.0134 (0.0123)	0.0267 (0.0201)
IB potential	0.0081 (0.0214)	-0.0086 (0.0216)	-0.0411*** (0.0118)	0.0029 (0.0065)	-0.0029 (0.0061)	-0.0024 (0.0136)
Share - syndicated loans	-0.1575* (0.0911)	-0.1330 (0.0928)	-0.0730 (0.1027)	-0.0597** (0.0261)	-0.0564** (0.0269)	-0.0893** (0.0413)
Share - underwriting	-0.0919 (0.0848)	-0.0767 (0.0835)	-0.0897 (0.1351)	-0.0419** (0.0177)	-0.0398** (0.0188)	-0.0995*** (0.0345)
Share - advisory	-0.1467 (0.1052)	-0.1314 (0.1025)	-0.1904 (0.1624)	0.0135 (0.0251)	0.0156 (0.0255)	0.0636 (0.0441)
Analyst FE	Yes	Yes	No	Yes	Yes	No
Firm FE	Yes	Yes	No	Yes	Yes	No
Year FE	No	Yes	No	No	Yes	No
Analyst-Year FE	No	No	Yes	No	No	Yes
Firm-Year FE	No	No	Yes	No	No	Yes
<i>N</i>	100,880	100,880	86,192	100,928	100,928	86,231
<i>R</i> ²	0.080	0.080	0.334	0.093	0.093	0.333

Panel A (broker characteristics). Bias is **mitigated** in more multicultural and higher-status broker environments:

- $\ln(\text{Nationalities}) \times \text{TrustBias}$ is negative and significant (about -0.20 for recommendations).
- $\text{BrokerHHI} \times \text{TrustBias}$ is positive and significant, consistent with *less diversity* (higher HHI) amplifying bias.
- $\text{Top10} \times \text{TrustBias}$ and $\ln(\text{BrokerSize}) \times \text{TrustBias}$ are negative, suggesting larger brokers attenuate bias.

Table XI
Cross-Sectional Differences in the Effect of Trust Bias

This table presents regression results for analyst recommendations. The dependent variable is shown above each column. Recommendation is the analyst recommendation, scaled from 1 (buy), strong buy, to 5 (hold), strong sell. Buy recommendation is a dummy taking the value of one if the recommendation is 1 (strong buy) or 4 (buy). Controls include three country dummies, ΔB (percentage ΔB percent), share, Δ (evaluated bias), share, Δ (evaluated bias), and share Δ (evaluated bias). The sample period is 1994 to 2018. Variables are defined in the Appendix. Heteroskedasticity-consistent standard errors, double-clustered by analyst-firm country pair and month, are shown in parentheses. Significance levels: *0.1, **0.05, ***0.01.

	Panel A: Broker characteristics					Buy rec.
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(\text{Nationalities}) \times \text{Trust bias}$	-0.203*** (0.0670)				-0.0774** (0.0231)	
$\text{Broker HHI} \times \text{Trust bias}$		1.050*** (0.3070)			0.2624*** (0.0881)	
$\text{Top 10} \times \text{Trust bias}$			-0.2832** (0.0921)		-0.0612* (0.0282)	
$\ln(\text{Broker size}) \times \text{Trust bias}$				-0.441*** (0.0461)		-0.0631** (0.0171)
Trust bias	1.050*** (0.0431)	0.2219 (0.1072)	0.0602*** (0.0172)	0.4118*** (0.0171)	0.1720** (0.0271)	0.4171*** (0.0161)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
First-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1,038,196	1,038,196	1,038,196	1,038,196	1,038,196	1,038,196
R ²	0.580	0.585	0.585	0.587	0.587	0.587

Table XI—Continued

	Panel B: Analyst characteristics					Buy rec.
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(\text{Time covered}) \times \text{Trust bias}$	0.0764*** (0.0170)				0.0030*** (0.0038)	
$\ln(\text{Experience}) \times \text{Trust bias}$		0.100** (0.0511)			0.0002*** (0.0020)	
Anti-globalization $\times$ Trust bias			1.440*** (0.0300)			1.433*** (0.0370)
Trust bias	0.1204 (0.2112)	-0.3859 (0.4344)	0.1110 (0.1701)	0.0795 (0.0832)	-0.1873 (0.1070)	-0.0603 (0.1197)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
First-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1,038,196	1,038,196	459,093	1,038,196	1,038,196	459,093
R ²	0.565	0.565	0.566	0.567	0.567	0.567

(Continued)

Table XI—Continued

	Panel C: Firm characteristics					Buy rec.
	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(\text{Market cap}) \times \text{Trust bias}$	0.1067 (1.0794)				0.1124*** (1.0861)	0.7070 (0.0270)
Market governance $\times$ Trust bias		0.1487*** (0.0721)			0.0627 (0.0778)	0.0473 (0.0381)
$\ln(\text{Market cap}) \times$ Trust bias			0.9428*** (0.3094)		0.0611 (0.2641)	0.2568* (0.1401)
$\ln(\text{Market cap}) \times$ Trust bias				0.1141*** (0.0201)		0.0177*** (0.0050)
Trust bias	0.7129*** (0.0012)	0.0224*** (0.1487)	0.0097* (0.0739)	0.0082*** (0.0041)	0.0084*** (0.0041)	0.2561*** (0.0012)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
First-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Analyst-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
N	121,498	1,038,196	964,270	1,038,196	1,038,196	964,270
R ²	0.574	0.585	0.581	0.586	0.577	0.586

Panel B (analyst characteristics). Bias is **stronger** for analysts with more time covering firms, more experience, and stronger anti-globalization attitudes: interactions are positive and significant.

Panel C (firm characteristics). The results are mixed for opacity proxies, but there is a robust positive interaction with size: $\ln(\text{MarketCap}) \times \text{TrustBias}$ is positive and significant. There is also evidence that institutional ownership and some opacity measures can interact positively with bias in certain specifications.

5 Short conclusion

The paper concludes that **cultural trust biases between nations have a statistically and economically meaningful impact on equity analysts' judgments**. These biases manifest primarily in recommendations and target prices, intensify when nationality is salient and when sentiment is negative, and respond to major political shocks (the European debt crisis and Brexit). Markets appear to partially recognize these biases (announcement reactions adjust), yet biased recommendations still have predictably lower information content for future monthly returns. At the same time, more accountable outputs such as EPS forecasts are not more optimistically biased and can even be more accurate, consistent with accountability disciplining bias in “hard” forecasts while leaving room for bias in more discretionary qualitative judgments.

6 Conclusion and intuition

6.1 Summary

This paper concludes that **cross-country cultural perceptions (measured by bilateral trust bias) systematically affect sell-side equity analysis in Europe**, and that this bias is **economically meaningful, state-dependent, and partially priced by markets**.

The evidence forms one coherent chain:

1. **Analyst recommendations exhibit cultural bias even under very demanding fixed effects.** In the baseline recommendation specification with **firm-month** and **analyst-month** fixed effects (so identification is within analyst and within firm each month), higher bilateral trust bias between an analyst’s country and a firm’s country predicts more optimistic calls (Table II). With full controls, the trust-bias coefficient is about 0.5916 for the 1–5 recommendation score, 0.2802 for the buy-indicator, and -0.1421 for the sell-indicator. Using the recommendation-sample standard deviation of trust bias ($\sigma \approx 0.148$, Table I), a one-standard-deviation increase in trust bias implies:

$$\begin{aligned}\Delta \text{Rec} &\approx 0.5916 \times 0.148 \approx 0.088, & \Delta \text{Pr}(\text{Buy}) &\approx 0.2802 \times 0.148 \approx 0.041, \\ \Delta \text{Pr}(\text{Sell}) &\approx -0.1421 \times 0.148 \approx -0.021.\end{aligned}$$

Thus, culturally-shaped beliefs shift recommendations by nontrivial amounts even after controlling for standard geography and investment-banking channels.

2. **Bias is stronger when nationality is more salient.** The trust-bias effect is amplified for “nationally salient” firms: the interaction $\text{Eponymous} \times \text{TrustBias}$ is positive and significant (e.g., ≈ 0.32 for recommendations and ≈ 0.12 for buy propensity), and the interaction $\ln(\text{MarketCap}) \times \text{TrustBias}$ is also positive (Table III). This implies that cultural bias is not uniform; it intensifies when nationality cues are more cognitively available.
3. **Bias strengthens in bad times and varies with sentiment.** The time-variation analysis shows that the effect of trust bias rises with pessimism and falls with consumer confidence (Figure 1 and Table IV). This is consistent with cultural stereotypes becoming more influential when uncertainty and stress are higher.
4. **Major political shocks shift cross-country biases in predictable directions.** During the European debt crisis, northern analysts become significantly more negative toward southern firms: $\text{Crisis} \times \text{NorthernAnalyst} \times \text{SouthernFirm}$ is around -0.374 to -0.426 for recommendation levels and around -0.217 to -0.247 for buy propensity (Table V; Figure 2). Around Brexit, U.K. analysts become much more home-biased after Article 50: $\text{Post} \times \text{British} \times \text{SameCountry}$ is around 0.48 to 0.70 for recommendation levels and around 0.17 to 0.34 for buy propensity (Table VI; Figure 3).
5. **Markets partially discount culturally biased recommendations, and bias reduces information content.** Recommendation announcement returns are consistent with partial discounting: for buy recommendations, higher trust bias predicts *weaker* positive CARs (e.g., ≈ -0.013 to -0.018 , Table VII), while for sell recommendations the negative reaction is stronger when the sell is more informative (active analysts / downgrade-to-sell subsamples, Table VII). In the monthly return regressions, interactions of recommendation-quintile dummies with trust bias indicate that **recommendations from more biased analyst sets are less useful for predicting subsequent returns at the extremes** (Table VIII).
6. **Bias is present in target prices but not in directional earnings-forecast bias.** Trust bias significantly increases target prices scaled by current price (e.g., coefficients from ≈ 0.08 up to ≈ 0.24 depending on target-price age, Table IX). In contrast, trust bias does not significantly increase directional earnings forecast bias (PMFB), but it is associated with *lower absolute forecast errors* (PMAFE) (Table X). This distinction implies that cultural bias primarily affects **discretionary/soft judgments** (recommendations and targets) rather than the most verifiable “hard” forecasts.

6.2 Why these patterns arise

The paper’s mechanism is best summarized as **culturally-shaped priors interacting with salience, uncertainty, and accountability**:

1. **Cultural trust acts like a belief/heuristic channel.** Bilateral trust bias captures how favorably (or unfavorably) people from one country perceive people from another, net of general “trustingness” and general “trustworthiness.” When an analyst evaluates a foreign firm, that perception can distort subjective assessments of management quality, governance, transparency, or strategic credibility, which naturally map into recommendation and valuation judgments.
2. **Salience amplifies bias through priming.** When nationality is made more salient (eponymous firms, “national champions,” very large firms), cultural categories become more cognitively accessible, so the same trust bias has a larger marginal effect (Table III).
3. **Bad times increase reliance on heuristics.** Under stress and uncertainty, analysts may rely more on coarse narratives and stereotypes as decision aids. This generates the observed state-dependence: the trust-bias slope is larger when pessimism is higher and consumer confidence lower (Figure 1; Table IV).
4. **Political shocks shift perceived out-group risk.** Events like the European debt crisis and Brexit change the narrative environment and perceived “out-group” risk, producing sharp cross-country divergence in analyst behavior (Tables V–VI; Figures 2–3).
5. **Accountability disciplines “hard” forecasts more than “soft” opinions.** Earnings forecasts are easy to evaluate ex post, which constrains directional bias. Recommendations and target prices are more discretionary and multi-dimensional, leaving more room for culturally-shaped judgment while still appearing justifiable.

6.3 Takeaway

Pursiainen shows that bilateral cultural trust bias systematically shifts analysts’ recommendations and target prices—especially when nationality is salient and during bad times or political shocks—and that markets partially discount these biased signals; the bias weakens the information content of recommendations, while more verifiable earnings forecasts show little directional bias, consistent with cultural priors influencing soft judgments under uncertainty but being disciplined by accountability for hard forecasts.